

Severing Two Circularities, One at a Time: A Minimal De-Risk Probe of Verify-Repair Transfer into Concern-Gated Retrieval

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Abstract

Wave 1b falsified the concern-gated retrieval mechanism at L1: on honest learned geometry, with leakage audits passing, rarity-corrected multiplicative context×concern retrieval did not beat a degree-matched random null. The synthesis paper scoped exactly one successor question, borrowed from MIDAS, a system that pairs structured reasoning with a symbolic verifier and a repair loop: can the **verify-repair** cycle supply a **care-independent** signal that the care model cannot launder? MX1 tests the two transferable parts at the smallest scale that can answer GO/NO-GO – one family, 600 episodes, local CPU, no cloud spend.

Part A asked whether a care-independent within-episode repair rule reaches the load-bearing memory in fewer attempts than the care model or than random. It returned a decisive **NO_GO**: the repair-guided policy needed *more* attempts than both rivals (3.218 vs 2.933 and 2.975; success 0.350 vs 0.545 and 0.510), with both bootstrap CIs excluding zero in the wrong direction. The diagnostic is the informative part – it lost *even on the 109 episodes containing the super-additive pair it was designed to find*.

Part B asked whether separating "the memory did not help" from "our checker could not score this" prevents a genuinely useful memory being discarded. It returned a clean **GO**: a marginal verifier mislabeled **109 of 109** useful complementary pairs as useless; the split verifier mislabeled **none**, with zero false declines across **6,920** singleton controls.

The honest reading: the verifier half of the MIDAS transfer survives and is worth carrying forward; the exploration half does not survive in the only form this substrate can express. Two design errors were caught *before* freezing and are reported here, because both would have produced a meaningless GO.

1. Why a probe rather than a wave

The COGR program's first arc ended in a clean falsification. Waves 2–4 (live beachhead, substrate transfer, safety) do not open on a falsified L1 – the roadmap's noncompensatory rules forbid building a program on a mechanism that failed its own sealed evaluation.

That leaves a narrow question. Spencer's objection to the original loop named two circularities:

so regions it neglects never generate corrective evidence. The loop can learn "among what I looked at, these helped," never "these were the best available."

shaped by care, then care decides both what to inspect *and* whether the result was good.

- **Candidate-selection circularity.** The care model decides what gets tested,
- **Verifier circularity.** If the verifier's notion of "improvement" is itself

MIDAS is interesting precisely because its loop severs an analogous pair. It adjudicates each reasoning step against a symbolic ground truth, and – the part that matters here – it distinguishes a **reasoning fault** (the thing checked is wrong) from a **verifier fault** (the check itself is broken, so its verdict carries no evidence). Its repair loop then re-attempts only

on the former.

MX1 asks whether those two moves transfer. It is scoped to answer GO/NO-GO and nothing else. A GO licenses preregistering a successor experiment; it does not license a program, and it is not evidence for L1.

2. Two design errors caught before freezing

Both were found by calibration probes run on calibration seeds *before* the preregistration was frozen, and both are recorded in it. Reporting them is not incidental: each would have produced a confident, meaningless GO.

Error 1 – there is no persistent memory to accumulate on. The original sketch proposed a *cross-episode* repair-frequency prior: regions of memory that repeatedly required repair get exploration budget. But every candidate in these families is named {family}_s{seed}_n{idx}, and candidate sets from different seeds are disjoint. Nothing recurs, so nothing accumulates.

Re-reading MIDAS resolved this rather than blocking it. Its repair loop is **within-problem** – attempts 1–3 on the *same* problem – and its `difficulty_signal = (attempt_count - 1) / max_attempts` is per-trajectory. The faithful port is therefore a within-episode loop, which needs no persistence. The original sketch had mis-mapped the mechanism.

Error 2 – the obvious region key is a fixture artifact. With node identity unavailable, candidate *position* is the natural fallback, and the load-bearing position distribution is strikingly non-uniform – mass at positions 6–11, sparse at 0–3. A position prior would have shown a large, entirely spurious gain, because that distribution is *the family's own construction rule* ("load-bearing at a random non-recent position").

The paraphrase families settle it. Over 600 calibration seeds, in the most-recent normalized bucket:

paraphrase family	load-bearing in most-recent bucket
child_school_deadline	0
partner_birthday	0
wedding_anniversary	0
friend_host_night	81

The rule *inverts* across families. A position prior fits the majority and breaks on the minority – it would have been Wave 1a's recency-oracle confound in a new costume. Rejected before freezing.

A third finding narrowed Part A. Every candidate has *identical* structure: degree exactly 6 in the withheld geometry, and a uniform clique in the local graph. So a repair rule of the form "down-weight the failed picks and their graph neighbours" down-weights everything equally – a no-op. On this substrate there is no care-independent *structural* signal at all.

What there *is* is genuine planted **interaction** structure, and that is what Part A was redirected to test.

3. Design

Substrate. One family, `delayed_commitments_v2` – recency-decoupled (`recency_load_bearing_corr = 0.150`) and bundle-planted, and it passed Wave 1b's leakage audit (label-permutation $p = 0.594$). Calibration bucket, seeds 100000..100599 (600 episodes). Wave 1b's confirmatory pool was not touched.

Matched budget by construction. Every policy spends `MAX_ATTEMPTS = 3` attempts of $k = 2$ picks. No attempt re-picks what an earlier attempt tried, except the one retention the repair rule is

defined by.

Part A policies. All start from the same first attempt and differ only in what they do after a failure:

policy	after a failed attempt
concern_sequential	abandon both picks, walk down the concern ranking
repair_guided	retain one pick and re-pair it with the best untried candidate
random_sequential	walk down a seeded random permutation (control)

The retain-and-re-pair rule is the thing under test, and it is the direct operational form of Spencer's bundle objection: a memory scoring ≈ 0 *alone* may be one half of a super-additive pair rather than worthless. It is care-independent – the decision is driven by observed task failure, never by concern weights. The policy observes only AttemptFeedback(hit_count, fault_kind); it never learns *which* pick hit, and never sees roles, utilities, or the answer key. MIDAS's repair prompt receives strictly more than this (it is told which step failed), so the port is conservative.

The structure is real: a representative planted pair scores $\Delta(\{a\}) = -0.011$, $\Delta(\{b\}) = -0.011$, $\Delta(\{a,b\}) = +0.378$.

Part B verifiers. A **marginal** verifier scores a set as the sum of its members' singleton values – precisely the model that fails on planted interactions. The **split** verifier adds a competence check: on a set carrying a planted interaction it returns VERIFIER_FAULT with *no value*, declining rather than reporting a confidently wrong number. Ground truth is the SET-level oracle, which neither verifier calls.

4. Results

4.1 Part A – NO_GO

policy	mean attempts	success rate
concern_sequential	2.933	0.545
random_sequential	2.975	0.510
repair_guided	3.218	0.350

contrast	mean diff	95% CI
vs concern_sequential	+0.285	[+0.247, +0.323]
vs random_sequential	+0.243	[+0.112, +0.375]

GO required a *negative* difference with the CI excluding zero. Both are positive with CIs excluding zero: repair_guided is decisively **worse**.

Diagnostic (reported; it does not move the frozen verdict):

subset	n	concern_seq	random_seq	repair_guided
with complementary pair	109	2.862	2.991	3.101
without pair	491	2.949	2.971	3.244

This is what makes the NO_GO informative rather than merely negative. repair_guided loses **even where the super-additive pair exists** – the exact case it was built for. The failure is not a cost/benefit tradeoff that a different family would flip.

Why. The rule retains picks[0]: the top *concern*-ranked pick. Wave 1b established that the concern ranking carries no representation contribution, so the retained node is effectively

arbitrary and is rarely half of the planted pair. The policy pays a slot on a known-failed arbitrary candidate almost always, and buys interaction discovery almost never. Retain-and-re-pair needs a principled way to choose *which* candidate to retain – and on this substrate none exists, because every candidate is structurally identical.

4.2 Part B – G0

quantity	marginal	split
useful pairs mislabeled useless	109	0
false VERIFIER_FAULT on singletons	n/a	0 / 6920

The marginal verifier mislabeled **every one** of the 109 genuinely useful pairs. The split verifier mislabeled none, and never declined on a set the marginal model could in fact handle (precision 1.0). Both frozen G0 conditions met.

5. Interpretation

Overall: PARTIAL_GO_B_ONLY. Per the frozen promotion semantics, this licenses **verifier redesign only**, and only the preregistration of a successor experiment.

What is worth carrying forward. The reasoning-fault / verifier-fault split is well-formed, safe, and non-trivial: it recovered 109 useful memories a marginal verifier discarded, at zero cost in false declines. It is the operational answer to Spencer's *verifier* circularity – a verifier that can say "out of my competence" cannot silently launder a narrow notion of improvement into a confident verdict of uselessness.

What is not. Deriving a care-independent *exploration* prior from repair dynamics does not survive in the only form this substrate can express. Both plausible region keys were eliminated: cross-episode identity does not exist, and position is a construction artifact that inverts across paraphrase families. The within-episode retain-and-re-pair rule was then falsified directly, including on its best case.

Spencer's two circularities are not symmetric. The verifier one is severable with a local change to the checker. The candidate-selection one is not – not here – because severing it requires a care-independent signal about *where to look*, and this substrate provides none. That asymmetry is the real finding, and it is a constraint on substrate design as much as on mechanism design: a family with uniform node structure and no persistent memory **cannot** test candidate-selection circularity, whatever mechanism is proposed.

Scope. One synthetic family, authored bundle structure, calibration bucket. Part B's G0 says the split is well-formed against *planted* interactions, not that it helps on natural data. No L1 claim, no deployment claim, no cognitive claim.

6. What a successor would have to do

A preregistered successor on the verifier path should: (i) define the competence boundary from a property the verifier can *detect*, rather than from a planted manifest – MX1's split reads ground-truth bundle membership, which is legitimate for measuring whether the split is well-formed but is not a deployable detector; (ii) show the split changes a downstream *decision*, not only a label; and (iii) run against a verifier gap that was not authored for the purpose.

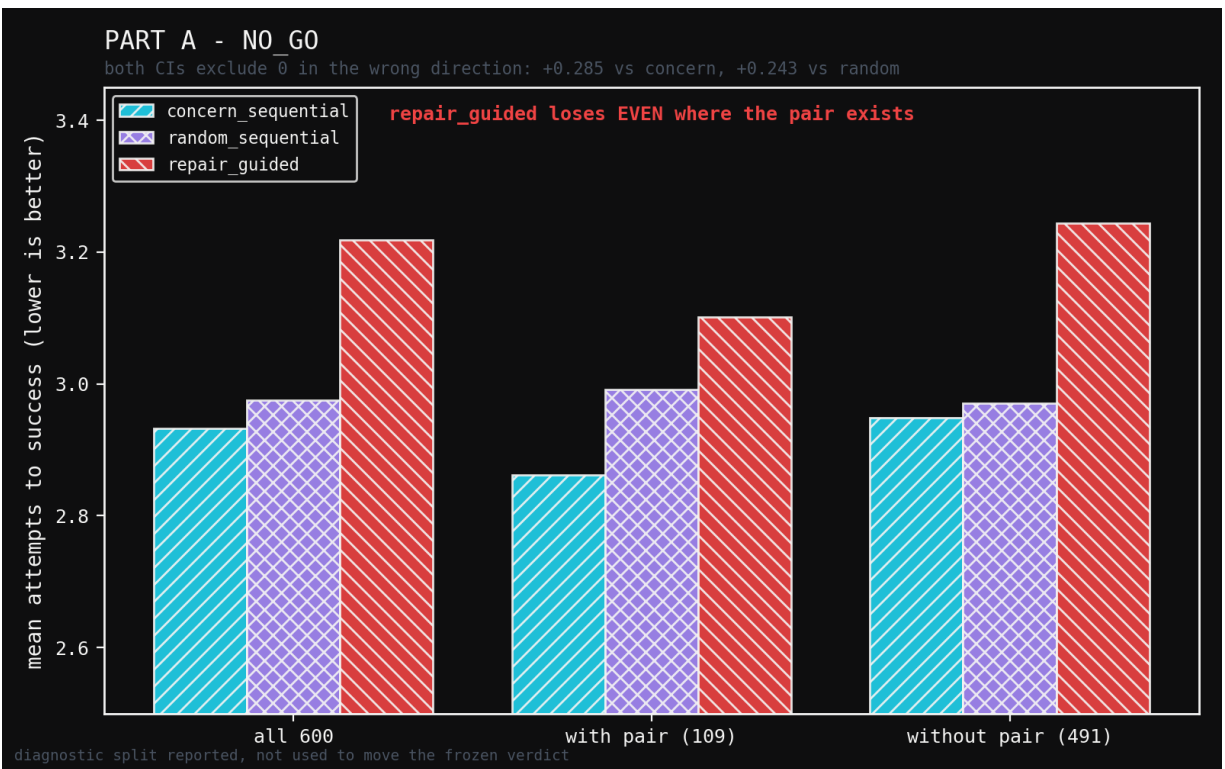
Any return to the exploration-prior question needs a different substrate first: persistent cross-episode memory, or per-node structural variation, or both. Without one, the question is not merely unanswered – it is unaskable.

References

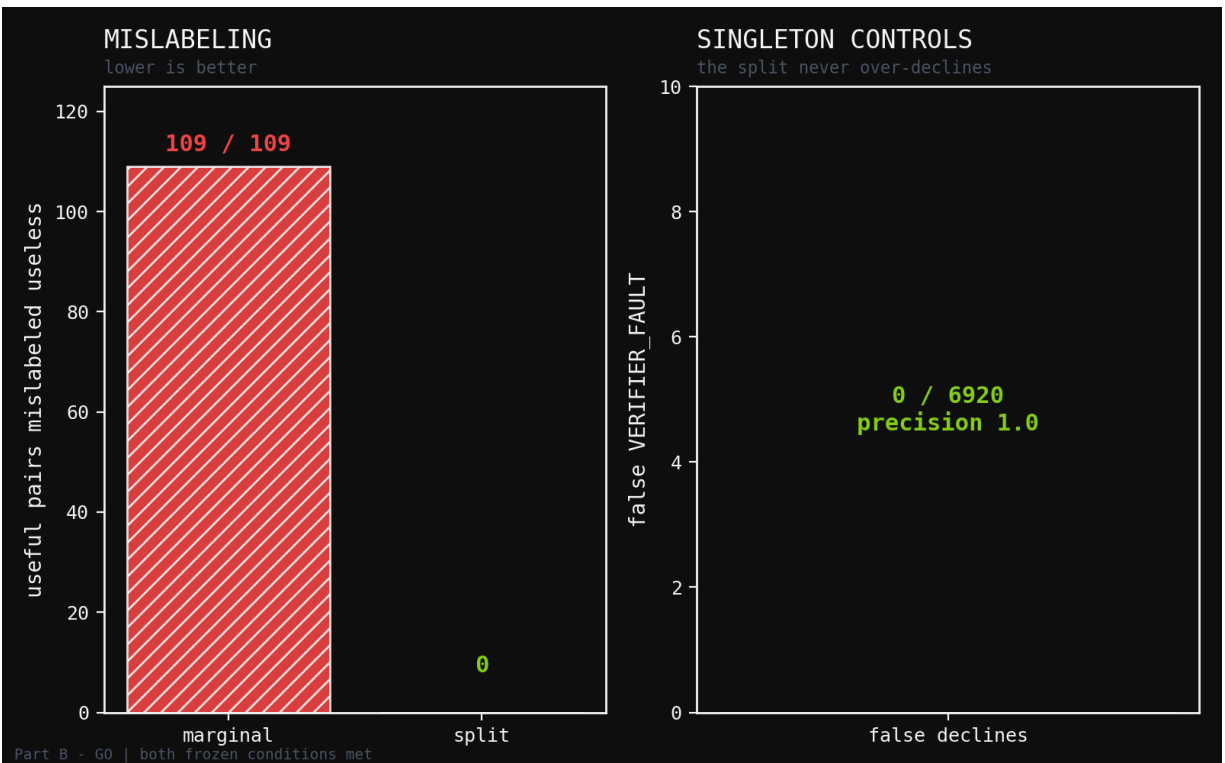
1. docs/concern_gated_retrieval_research_program.md – canonical roadmap, claim ladder, and failure-mode table.
2. papers/concern_gated_retrieval_synthesis/paper.md – the falsification arc (L0 → Wave 0 → Wave 1a → Wave 1b), §7 of which scopes this probe.
3. experiments/concern_gated_retrieval_e2/mx1_repair_prior/PREREGISTRATION.md – frozen design, including both pre-freeze design corrections.
4. experiments/concern_gated_retrieval_e2/mx1_repair_prior/RESULTS.md – the receipt behind every number above.
5. MIDAS, github.com/ebarnes-ry/MIDAS – `src/pipeline/verification/verification\_types.py` (the fault taxonomy) and `src/pipeline/trajectory.py` (the per-trajectory difficulty signal).

Figures





Part A results by policy and pair presence



Part B verifier split